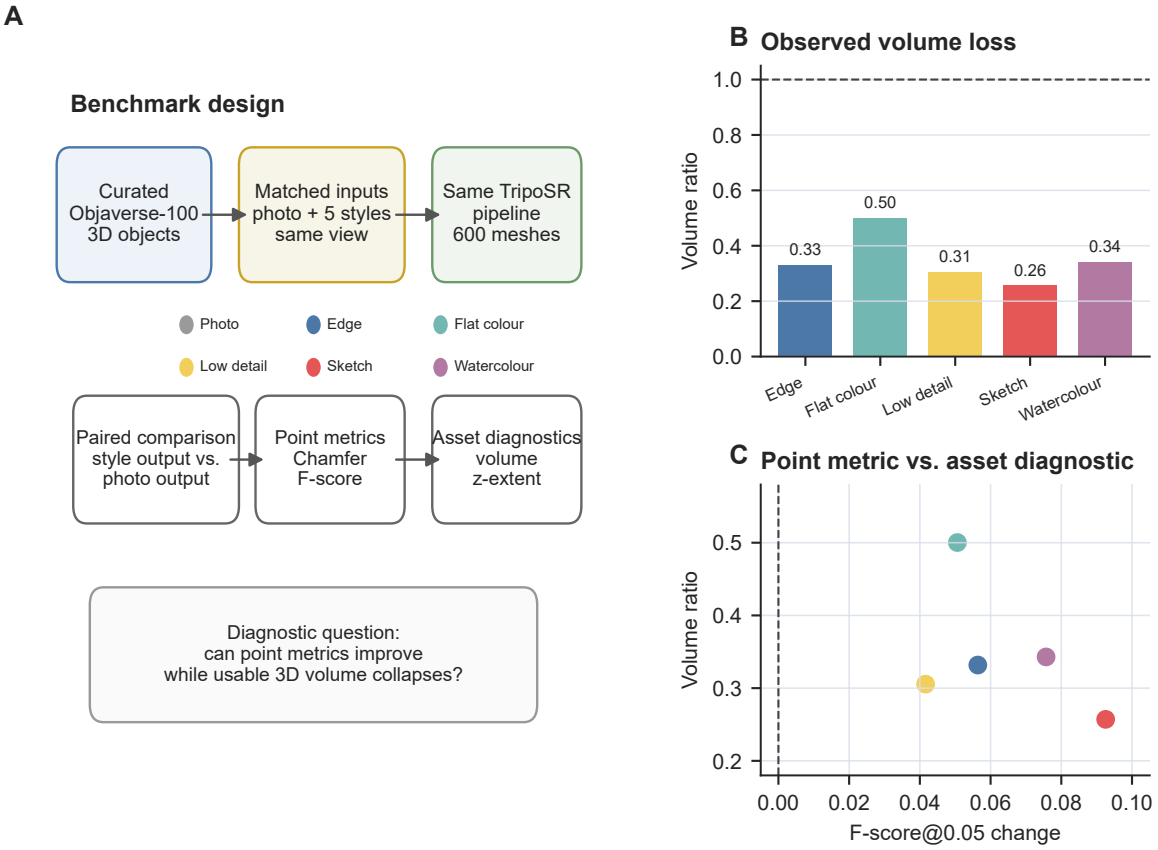

A
Benchmark design
Curated Objaverse-100 3D objects
Matched inputs photo + 5 styles same view
Same TripoSR pipeline 600 meshes
Photo
Edge
Flat colour
Low detail
Sketch
Watercolour
Paired comparison style output vs. photo output
Point metrics Chamfer F-score
Asset diagnostics volume z-extent
Diagnostic question: can point metrics improve while usable 3D volume collapses?
B
Observed volume loss
Volume ratio
1.0
0.8
0.6
0.4
0.2
0.0
0.33
0.50
0.31
0.26
0.34
Edge
Flat colour
Low detail
Sketch
Watercolour
C
Point metric vs. asset diagnostic
Volume ratio
0.5
0.4
0.3
0.2
0.00
0.02
0.04
0.06
0.08
0.10
F-score@0.05 change